

Project Proposal: SolarPulse Analytics

Renewable Energy Data Analytics & Predictive Modeling for Sustainability

1. Project Title

Renewable Energy Data Analytics & Predictive Modeling for Sustainability

2. Executive Summary (The Scenario)

Our team is developing **SolarPulse Analytics** to address the instability of solar power production. Solar plants suffer from output fluctuations due to weather variables (clouds, heat, and dust). By building an integrated data system that links weather sensors with plant production using **SQL** and **Python**, and applying **Machine Learning**, we aim to transform raw data into precise forecasts. This helps grid managers prevent energy waste and ensure a steady power supply.

3. Problem Statement

- **Production Volatility:** Solar output is unstable due to rapid weather changes, making it unreliable for direct grid dependence.
 - **Planning Uncertainty:** Without accurate forecasts, it is difficult for national grids to balance load and demand.
 - **Hidden Inefficiency:** Environmental factors like "Module Overheating" or "Dust" reduce efficiency without being detected by traditional monitoring.
-

4. Project Objectives

- **Unified Data Integration:** Architecting a relational database (**SQL**) to sync power production with meteorological variables.
- **AI-Driven Forecasting:** Building a **Machine Learning** model to forecast future energy yield based on weather inputs.
- **Anomaly Detection:** Identifying performance gaps to pinpoint maintenance needs (e.g., panel cleaning).
- **Strategic Visualization:** Delivering an interactive **Power BI** dashboard for real-time decision support.

5. Technical Methodology (The Workflow)

Phase 1: Data Engineering & Analytics (SQL & Python)

- **Cleaning & Audit:** Initial audit in **Excel**, followed by **Python (Pandas)** to handle missing values and outliers.
- **Data Merging (SQL):** Importing CSVs into a relational database and using **SQL JOINS** to link Generation and Weather tables via the DATE_TIME key.
- **Exploratory Analysis (EDA):** Visualizing correlations (e.g., the "Heat Penalty" where high temperatures lower panel efficiency).

Phase 2: AI Integration & Modeling (Machine Learning)

- **Model Building:** Implementing **Supervised Learning** (e.g., **Linear Regression** and **Random Forest**) using **Scikit-Learn**.
- **The AI Task:** The model will process Irradiation and Temperature to output a **Predicted Power Yield**.
- **Evaluation:** Comparing AI forecasts with actual production to measure accuracy (R^2 and MAE).

Phase 3: Final Visualization (Power BI)

Creating an interactive dashboard featuring:

- **Historical Trends:** Daily and total yield tracking.
- **Actual vs. Predicted:** A visual comparison of AI accuracy.
- **Maintenance Alerts:** Indicators for underperforming inverters.

6. Risk Assessment & Mitigation Plan

Identified Risk	Impact	Mitigation Strategy (Solution)
Data Gaps (Missing sensor readings)	High	Use Data Imputation (Mean/Interpolation) in Python to fill gaps.
Model Overfitting	Medium	Use Train-Test Split and Cross-Validation to ensure general accuracy.
Sensor Inaccuracy	Medium	Perform Outlier Detection during the cleaning phase to remove noise.

7. Key Performance Indicators (KPIs)

- **Model Accuracy:** Achieve an **R^2 Score > 90%** for energy prediction.
 - **Error Minimization:** Reduce **Mean Absolute Error (MAE)** to acceptable industrial limits.
 - **Insight Discovery:** Successful identification of the temperature threshold that affects efficiency.
 - **Dashboard Functionality:** 100% integration between SQL/Python outputs and Power BI visuals.
-

8. Project Plan (Gantt Chart - 4 Weeks)

- **Week 1: Foundations** – Data Collection, SQL Database Setup, and Initial Excel Audit.
 - **Week 2: Engineering** – Data Cleaning, SQL Joining, and EDA in Python.
 - **Week 3: Intelligence** – Machine Learning Model Training, Testing, and Comparison.
 - **Week 4: Delivery** – Power BI Dashboard Design and Final Presentation Preparation.
-

9. Team Roles & Responsibilities (4-Person Team)

- **Member 1 (Data Architect):** SQL Database management, data integrity, and Excel documentation.
 - **Member 2 (Data Analyst):** Lead for Python-based EDA and discovering environmental insights.
 - **Member 3 (ML Engineer):** Responsible for building, tuning, and evaluating AI models.
 - **Member 4 (BI Developer & Coordinator):** Designing the final Power BI Dashboard and project coordination.
-

10. Impact & Sustainability

This project aligns with **UN Sustainable Development Goal #7 (Clean Energy)**. By turning raw data into **Predictive Intelligence**, our team aims to demonstrate how modern data tools can reduce carbon footprints and lead the way to a smarter, greener planet.